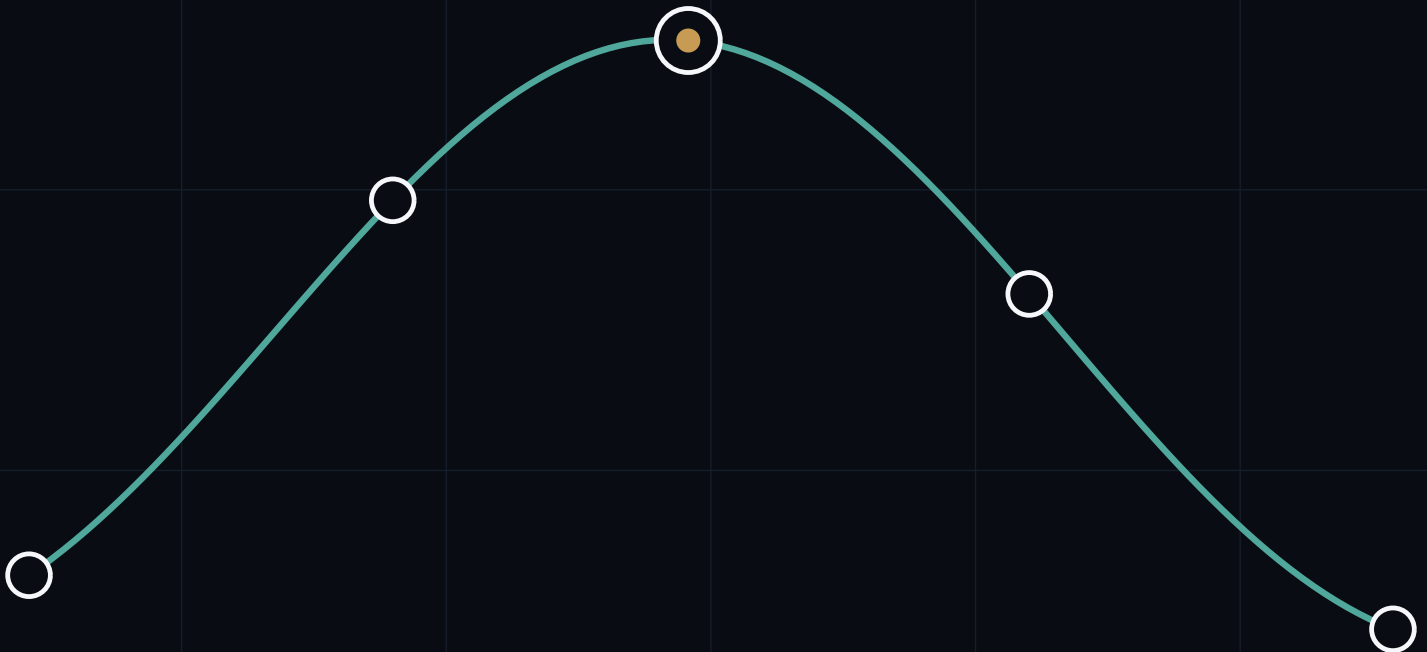
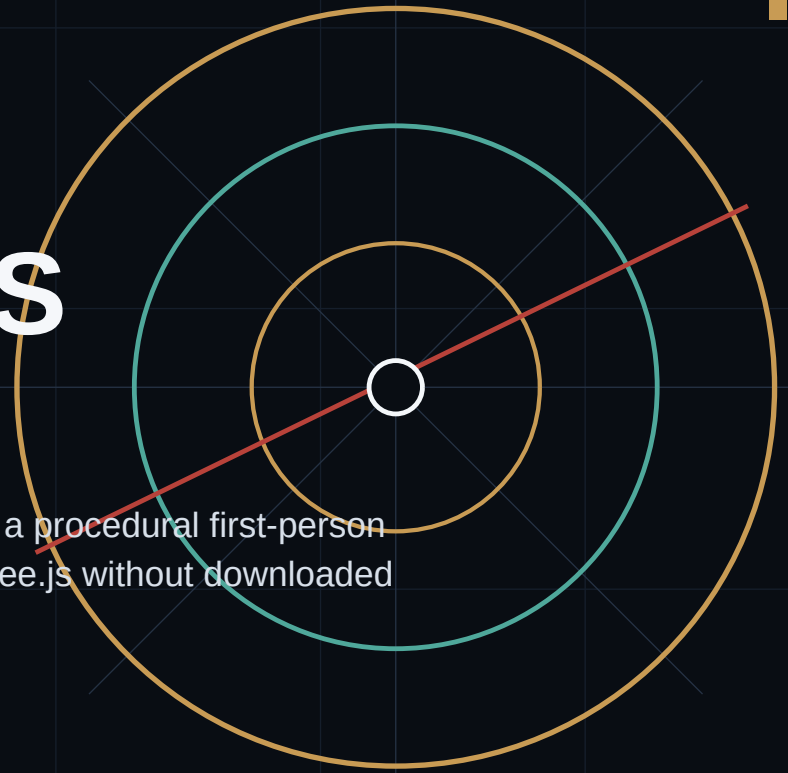


FULL MULTI-AGENT PRODUCTION PROMPT

THE GLASS OSSUARY

A complete evidence-graph prompt for a procedural first-person mystery horror vertical slice built in Three.js without downloaded assets.



COPY-PASTE PROMPT + AGENT CARDS

Product contract, game systems, subagent topology, deterministic simulation, procedural art grammar, encounter and puzzle design, QA, performance, provenance, repair, rollback, and release gates.

THREE.JS r185 (0.185.0) RUNTIME
12-16 MINUTE MYSTERY HORROR SLICE
ONE INQUEST / THREE APPARITIONS / ONE BOSS
DIEGETIC OPTICS, PHONOGRAPH & FERROTYPE
INDEPENDENT EVIDENCE AND CRITICS

00

DEPLOYMENT DECLARATION & GENRE PREMISE

MYSTERY ON HORROR SLICE

An executable multi-agent production contract for a 12-to-16-minute investigative psychological horror vertical slice set in a storm-lashed coastal bone-glass observatory.

Forensic Investigation Meets Acoustic Horror

Every scare and revelation is governed by deterministic acoustic decibel radii, refractive prism geometry, and a 6-node Deduction Board rather than scripted jump-scare cutscenes.

100% Source-Generated in Three.js r185

All ribbed basalt vaults, Fresnel glass lenses, tallow lantern shaders, wax-cylinder EVP synthesis, and apparition silhouettes are compiled from code.

Dual Evidence Regimes (A-PinnedBrowser & B-CrossPlatform)

Every clue state, deduction branch, 60 Hz stealth tick, and 16-bit PCM hydrophone render is verified before any node advances.

PUBLICATION STRUCTURE

Contents & Multi-Agent Execution Map

Read the Mystery Horror game contract and 10-beat investigation route first, then the 15-node Evidence Graph, diegetic tool math, and deterministic QA gates. Sections 05 and 06 provide the copy-paste prompts.

Section	Pages	Title & Normative Deliverable
Section 00	pp. 02-04	Deployment declaration, Mystery Horror design laws, and v2.0 basis
Section 01	pp. 05-10	Game contract, 5 coastal ossuary spaces, and 10-beat inquest route
Section 02	pp. 11-15	15-node Evidence Graph, specialist write isolation, and provenance
Section 03	pp. 16-23	Diegetic tools, 60 Hz stealth/sanity math, apparitions, and boss
Section 04	pp. 24-27	16 deterministic replay scenarios (rep-01..16), perf & critic gates
Section 05	pp. 28-30	Full copy-paste Master Orchestrator prompt for The Glass Ossuary
Section 06	pp. 31-34	Seven specialist subagent prompt cards (combat/stealth to critic)
Appendix A	pp. 35-36	Draft 2020-12 JSON Schemas and 20 clickable primary references

Protagonist: Clara Vane (The Acoustic Archivist)

100 Composure / 100 Oil / 0-100 Exposure

Sent to Saint-Vane Tidal Observatory to investigate the vanished lighthouse commission using three optical and acoustic instruments.

Core Loop: Observe -> Record -> Triangulate -> Deduce

Zero HUD Clutter

Inspect micro-fractures with the Brass Loupe, phase-invert room drones with the Wax Phonograph, expose spectral occlusions with Ferrotypes Plates, and lock hypotheses on the Evidence Board.

EVIDENCE GRAPH v2.0 INVARIANT // NAVIGATION / CONTENTS

Deterministic Gate Verification for Contents & Multi-Agent Execution Map

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

ANTI-SLOP HORROR & DEDUCTION DISCIPLINE

Eight Non-Negotiable Mystery Horror Laws

Procedural horror fails when darkness is an unlit black screen or when mysteries solve themselves through linear dialogue prompts. The Glass Ossuary enforces eight mechanical laws.

Law 01: Readable Darkness (Luminance Floor ≥ 1.8 cd/m²)

Darkness uses wet abyssal slate (#070B10) and basalt (#16202A) with specular edge rim-lighting from tallow amber (#C89B54) and silver-halide cyan (#4FA89B). Never crush frame histograms to pure black.

Palette Lock

Law 02: Acoustic Raycast Honesty

Apparitions hunt by deterministic footstep, lantern-shutter, and phonograph decibel envelopes (-48 dBFS silent crouch to -12 dBFS sprint splash), never by omniscient player coordinates.

60 Hz dB Budget

Law 03: Earned Forensic Deduction

Clues gathered in the 5 spaces feed the Stripping Room Evidence Board. Locking a hypothesis alters tool parameters and boss vulnerability windows.

6-Node Contradiction Matrix

Law 04: Zero Cheap Jump-Scare Stingers

Tension comes from sub-bass standing waves, glass-rib sympathetic resonance, and spatialized Web Audio binaural panning capped at -1.0 dBTP.

EBU R128 -16 LUFS

Swatch Hex	Material Role	TSL Surface Shader & Roughness Contract
#070B10	Abyssal Tide Slate	Wet tidal slate & shadow baseline; roughness 0.18-0.32 with micro-ripples
#16202A	Salt-Frosted Basalt	Vaulted masonry ribs & piers; procedural salt-crust Voronoi normal map
#C89B54	Tallow Amber	Diegetic lantern flame, brass loupe housing, and safe acoustic nodes
#4FA89B	Silver-Halide Cyan	Ferrotypes latent trails, prism caustics, and phase-cancel resonance
#B8423A	Arterial Rust	Contradiction alerts, high-exposure distortion, and apparition hazard

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 00 / DESIGN LAWS

Deterministic Gate Verification for Eight Non-Negotiable Mystery Horror Laws

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01

WORLD ARCHITECTURE & TEN-BEAT INQUEST ROUTE

GAME CONTRACT & ROUTE

Five interconnected coastal observatory spaces, one overarching forensic inquest (case_saint_vane), three investigative instruments, three stalker apparitions, and a two-phase acoustic climax.

Five Spaces, Zero Loading Screens

Tidewater Causeway -> Caretaker's Stripping Room -> Refraction Gallery -> Submerged Crypt -> The Glass Ossuary.

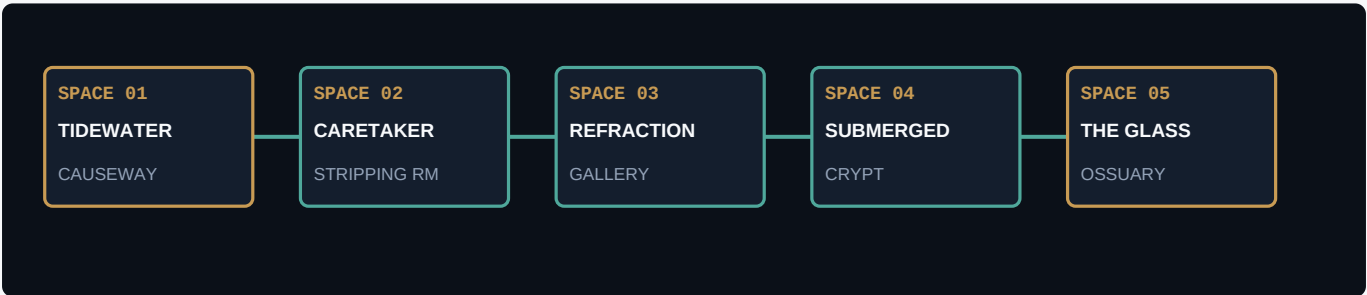
Ten Authored Investigation & Horror Beats

Every beat introduces one forensic mechanic, verifies it against an environmental obstacle or apparition, and records a deterministic state hash.

FIVE AUTHORED ARCHITECTURAL SPACES

Spatial Topology of Saint-Vane Observatory

All five spaces stream within a single contiguous Three.js scene graph separated by acoustic airlocks and refractive prism shutters so draw calls remain <= 280 and triangles <= 460,000.



Space ID	Architectural Signature	Key Forensic Clue / State Gate	Primary Hazard
Tidewater Causeway	Basalt sea-wall & storm lantern pier	Docking chain & oil canister	Gale wind & footing drift
Stripping Room	Oak inquest desk & 6-node Evidence Board	Caretaker Moreau wax cylinder	Safe hub / board lock
Refraction Gallery	3 rotating Fresnel glass rings & catwalks	Tide Ledger Fragment (optical)	Glass Septum Watcher
Submerged Crypt	Knee-deep tidal vaults & sluice gates	Hydrophone Cylinder (acoustic)	Drowned Chorister
The Glass Ossuary	Ribbed bone-glass acoustic cathedral	Saint-Vane Master Seal	The Choir in the Glass

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / SPATIAL GRAPH

Deterministic Gate Verification for Spatial Topology of Saint-Vane Observatory

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AUTHORED PROGRESSION SEQUENCE

Ten-Beat Investigation & Horror Route

The inquest state machine (case_saint_vane) advances monotonically through ten authored beats. Skipping a proof gate or mutating state outside the active beat is rejected by the orchestrator.

Beat	Canonical Name	Space	Forensic Task & State Transition
01	Causeway Landfall	Tidewater Causeway	Calibrate lantern shutter, crouch dB meter, reach airlock
02	The Sealed Inquest	Stripping Room	Play Moreau cylinder; case_saint_vane -> INQUEST_OPENED
03	Ferrotypes Calibration	Darkroom Corridor	Expose latent handprints with Ferrotypes Plate; evade Mire Listener
04	Refraction Gallery	Refraction Gallery	Use Brass Loupe to trace hairline fractures past Septum Watcher
05	Prism Triangulation	Refraction Gallery	Align 3 Fresnel rings; acquire Tide Ledger (OPTICAL_PROOF)
06	Submerged Crypt	Submerged Crypt	Phase-cancel Drowned Chorister hum; acquire Hydrophone Cylinder
07	Inquest Board Deduction	Stripping Room	Lock 1 of 3 hypotheses (lens_sabotage / tidal_quarantine / acoustic_calling)
08	Ossuary Unsealing	Ossuary Threshold	Resonate bone-glass lock with chosen hypothesis (OSSUARY_UNSEALED)
09	The Choir in the Glass	The Glass Ossuary	Two-phase acoustic/optical boss confrontation (CHOIR_SILENCED)
10	Publish or Submerge	The Glass Ossuary	Choose VERDICT_PUBLISH or VERDICT_SUBMERGE; persist save

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / TEN-BEAT ROUTE

Deterministic Gate Verification for Ten-Beat Investigation & Horror Route

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

FORENSIC STATE MACHINE & HYPOTHESIS LOCK

The Six-Node Evidence Board & Three Hypotheses

Unlike RPG level grinding, progression in The Glass Ossuary is forensic: linking collected clues on the Stripping Room Evidence Board unlocks one of three mutually exclusive hypotheses at Beat 07.

```
// Canonical Inquest State Progression (case_saint_vane)
type InquestState =
| 'NOT_STARTED'           // Beat 01: Causeway arrival
| 'INQUEST_OPENED'       // Beat 02: Caretaker Moreau wax cylinder played
| 'FERROTYPE_CALIBRATED' // Beat 03: Latent salt-print cipher exposed
| 'OPTICAL_PROOF_ACQUIRED' // Beat 05: Tide Ledger Fragment recovered
| 'ACOUSTIC_PROOF_ACQUIRED' // Beat 06: Hydrophone Cylinder recovered
| 'HYPOTHESIS_LOCKED'    // Beat 07: lens_sabotage | tidal_quarantine | acoustic_calling
| 'OSSUARY_UNSEALED'     // Beat 08: Bone-glass portal opened
| 'CHOIR_SILENCED'       // Beat 09: The Choir in the Glass defeated
| 'VERDICT_PUBLISH'      // Beat 10A: Transmit inquest cylinder to mainland
| 'VERDICT_SUBMERGE'     // Beat 10B: Open sea-cocks to submerge the ossuary
| 'COMPLETE';
```

Hypothesis ID	Canonical Name	Mechanical Modifier (Integer 60 Hz)	Trade-off / Penalty
lens_sabotage	Lens Sabotage	Loupe focus distance +40%; Ferrotypes stun +30 ticks	Lantern oil burn rate +25%
tidal_quarantine	Tidal Quarantine	Wading footstep dB -6 dBFS; Composure regen +20%	Phonograph windup +6 ticks
acoustic_calling	Acoustic Calling	Phonograph cancel window +24 ticks (114 total)	Exposure gain on hit +20%

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / DEDUCTION BOARD

Deterministic Gate Verification for The Six-Node Evidence Board & Three Hypotheses

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DIEGETIC KIT, CONSUMABLES & TESTIMONY

Archivist Kit, Consumables & Wax-Cylinder Transcripts

Every item in Clara Vane's satchel has a physical 3D procedural mesh, a deterministic tick cost, and a persistence entry in the local save schema.

Item ID	Canonical Name	Category	Deterministic Function & Stack Limit
brass_loupe	Split-Diopter Brass Loupe	Instrument	Reveals prism angles & micro-etchings within 3.2 m
wax_phonograph	Wax-Cylinder Phonograph	Instrument	Records room harmonics & emits 90-tick phase-cancel wave
ferrotype_plate	Silver-Salt Ferrotype Plate	Instrument	18-tick charge, 6-tick UV flash; stuns apparitions 150 ticks
smelling_salts	Camphor Smelling Salts	Consumable (2)	Restores +40 Composure and clears -25 Exposure over 30 ticks
whale_oil_vial	Refined Spermaceti Vial	Consumable (3)	Restores +50 Lantern Oil over 45 ticks (vulnerable to cancel)
tide_ledger	Tide Ledger Fragment	Key Proof	Optical proof from Refraction Gallery; unlocks Board Node B
hydro_cylinder	Hydrophone Cylinder	Key Proof	Acoustic proof from Submerged Crypt; unlocks Board NodeD

Caretaker Moreau (Recorded Testimony Contract)

3 Cylinder Tracks / Deterministic Subtitles

All narrative voice is delivered through procedural formant-filtered Web Audio speech tones paired with time-coded subtitles and zero external audio files.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / INVENTORY & DIALOGUE

Deterministic Gate Verification for Archivist Kit, Consumables & Wax-Cylinder Transcripts

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

SCOPE DISCIPLINE & NON-GOALS

Explicit Exclusions Protecting Slice Quality

To guarantee a finished, atmospheric 12-to-16-minute mystery horror experience, the following scope traps are strictly forbidden.

- No Firearms or Melee Weapon Combat

Pure Investigative Horror

Clara Vane is an acoustic archivist, not a soldier. Apparitions are evaded through acoustic stealth, misdirected with phonograph lures, or stunned via Ferrotpe flash and prism alignment.
- No Procedural Maze Randomization Across Same Seed

Authored Spatial Tension

Room geometry, sightlines, and clue placements are compiled deterministically from fixed seeds so replay scenarios (rep-01...rep-16) remain bit-exact.
- No Unskippable Cutscenes or Screen-Space Blood Splatters

Diegetic Feedback Only

Composure loss and Exposure buildup manifest as subtle lens astigmatism, brass lantern flicker, and binaural heartbeat harmonics, with a Reduced-Motion toggle.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / SCOPE BOUNDARIES

Deterministic Gate Verification for Explicit Exclusions Protecting Slice Quality

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

02

EVIDENCE GRAPH TOPOLOGY & WRITE ISOLATION

AGENT GRAPH & AUTHORITY

The 15-node directed cyclic Evidence Graph governs every build, verification, critic review, and rollback decision for The Glass Ossuary.

Single Write Owner Per File Path

Six specialist builders (combat_gameplay, world_quest, procedural_art_vfx, procedural_audio, ui_hud_accessibility, qa_perf_playwright) plus read-only independent_critic.

Maximum 3 Repair Iterations Before Rollback

If any P0/P1 defect fails to converge after 3 repair loops at N13_REPAIR_ROUTER, git worktree resets to accepted_baseline_commit.

CANONICAL 15-NODE PRODUCTION GRAPH

Node-by-Node Gate Contracts (N00 to N14)

Every node in the Evidence Graph emits schema-validated JSON artifacts and requires exit code 0 before downstream nodes unlock.

Node ID	Owner Role	Primary Deliverable & Gate Verification
N00_BRIEF	orchestrator	Freeze 12-16 min Mystery Horror contract & 5-color palette
N01_CONTRACTS	orchestrator	Validate task-packet, defect-record, run-manifest schemas
N02_SCAFFOLD	qa_perf_playwright	60 Hz integer clock, seeded PRNG, WebGPURenderer boot
N03_ASSET_COMPILER	procedural_art_vfx	Bone-glass ribs, Fresnel lenses, lantern & loupe meshes
N04_WORLD_GRAPH	world_quest	5 coastal spaces, acoustic portals, case_saint_vane graph
N05_RENDER_PIPELINE	procedural_art_vfx	TSL caustics, volumetric fog, compileAsync prewarm
N06A_GAMEPLAY_COMBAT	combat_gameplay	60 Hz stealth/exposure state machine, 3 apparitions, boss
N06B_AUDIO_SYNTH	procedural_audio	Wax phonograph, hydrophone EVP, -16 LUFS OfflineAudioContext
N07_UI_HUD_A11Y	ui_hud_accessibility	Evidence Board UI, subtitles, high-contrast & reduced-motion
N08_TELEMETRY_HARNESS	qa_perf_playwright	Wire window.__rpgReady and window.__rpgQA hooks
N09_REPLAY_RUNNER	qa_perf_playwright	Execute rep-01..rep-16; verify state, audio, geometry hashes
N10_PERF_GATE	qa_perf_playwright	Verify P50 <= 8.3 ms, P95 <= 16.6 ms, P99 <= 22.0 ms
N11_VISUAL_AUDIO_CRITIC	independent_critic	Order-reversed blind rubric review of frames & audio
N12_PROVENANCE_AUDIT	independent_critic	Confirm 0 external network requests & 0 downloaded assets
N13_REPAIR_ROUTER	orchestrator	Route DefectRecords (max 3 loops) or rollback worktree
N14_RELEASE_CANDIDATE	orchestrator	Emit signed run-manifest.json with cost_ledger

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / 15-NODE TOPOLOGY

Deterministic Gate Verification for Node-by-Node Gate Contracts (N00 to N14)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DISJOINT WRITE BOUNDARIES

Specialist File-Ownership Isolation Matrix

No specialist agent may modify files outside its allowed_paths or touch schemas/, orchestration/, or tests owned by QA.

Specialist Agent	Allowed Write Paths	Forbidden Paths
combat_gameplay	src/stealth/, src/apparitions/, src/tools/	src/render/, src/audio/, src/ui/, schemas/
world_quest	src/world/, src/inquest/, src/save/	src/stealth/, src/render/, src/audio/, schemas/
procedural_art_vfx	src/render/, src/procedural/, src/materials/	src/stealth/, src/audio/, src/ui/, schemas/
procedural_audio	src/audio/, tests/audio/	src/stealth/, src/render/, src/ui/, schemas/
ui_hud_accessibility	src/ui/, src/board/, src/input/	src/stealth/, src/render/, src/audio/, schemas/
qa_perf_playwright	src/telemetry/, tests/playwright/, scripts/	src/stealth/, src/render/, src/audio/, schemas/
independent_critic	evidence/ (Read-Only across all src/)	src/, schemas/, orchestration/, prompts/

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / OWNERSHIP MATRIX

Deterministic Gate Verification for Specialist File-Ownership Isolation Matrix

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REGIME A & REGIME B VERIFICATION

Deterministic Hashing for Optics, Audio & Stealth

The Glass Ossuary uses integer 60 Hz simulation ticks, 1e-5 quantized geometry buffers, and 16-bit PCM deadband quantization for OfflineAudioContext hydrophone renders.

```
// Regime A (A-PinnedBrowser) Hash Inputs for The Glass Ossuary
interface OssuaryStateSnapshot {
  tick: number; // Fixed 60 Hz integer tick
  inquest_state: InquestState; // case_saint_vane state
  composure: number; // 0..100 integer-scaled (x100)
  lantern_oil: number; // 0..10000 centi-units
  exposure: number; // 0..100 integer-scaled
  active_hypothesis: 'none' | 'lens_sabotage' | 'tidal_quarantine' | 'acoustic_calling';
  prism_angles_deg: [number, number, number]; // Integer degrees 0..359
  apparition_states: Array<{ id: string; node: number; alert_db: number; stunned_ticks: number }>;
}
```

Zero External Asset Provenance Gate (N12)

`external_network_requests === 0`

Playwright intercepts all page.on('request') events during rep-01..rep-16. Any fetch for .glb, .gltf, .png, .jpg, .ktx2, .mp3, .ogg, .wav, or .woff2 fails the build immediately.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / DETERMINISM REGIMES

Deterministic Gate Verification for Deterministic Hashing for Optics, Audio & Stealth

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WEBGPU + WEBGL2 FALLBACK ARCHITECTURE

Refractive Glass, Volumetric Mist & Pre-Warming

All bone-glass ribs and Fresnel lenses use TSL NodeMaterial transmission and angular dispersion with pre-warmed shader pipelines.

```
// Mandatory Shader Pre-Warming Before First Interactive Frame (N05 / N10)
const renderer = new THREE.WebGPURenderer({ antialias: true, forceWebGL: useFallback });
await renderer.init();
await renderer.compileAsync(ossuaryScene, archivistCamera);
await postProcessing.renderAsync(); // Eliminates first-flash Ferrottype stall
window.__rpgReady = true;
```

Budget Metric	Target Ceiling	Enforcement Node
Frame Time P50 / P95 / P99	<= 8.3 ms / <= 16.6 ms / <= 22.0 ms	N10_PERF_GATE (1920x1080)
Draw Calls Per Frame	<= 280 draw calls (Instanced/Batched)	N05_RENDER_PIPELINE & N10
Active Scene Triangles	<= 460,000 triangles	N03_ASSET_COMPILER & N10
GPU Memory Footprint	<= 384 MB total GPU allocation	N10_PERF_GATE

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / THREE.JS r185 PIPELINE

Deterministic Gate Verification for Refractive Glass, Volumetric Mist & Pre-Warming

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03

INVESTIGATION TOOLS, STEALTH MATH, APPARITIONS & BOSS

SYSTEMS & HORROR MECHANICS

Exact 60 Hz frame timings for diegetic instruments, acoustic detection envelopes, three apparition archetypes, and the two-phase climax against The Choir in the Glass.

Three Diegetic Instruments

Split-Diopter Brass Loupe, Wax-Cylinder Phonograph, and Silver-Salt Ferrottype Plate.

Three Stalker Apparitions + One Two-Phase Boss

Mire Listener, Glass Septum Watcher, Drowned Chorister, and The Choir in the Glass.

DETERMINISTIC 60 HZ ACTION TIMINGS

Diegetic Instrument & Evasion Frame Table

Every player action in The Glass Ossuary is governed by integer 60 Hz tick counts (1 tick = 16.6667 ms) so replay traces remain deterministic.

Action	Startup	Active Window	Recovery	Total	Cost & Mechanical Effect
Lantern Shutter	6 ticks	Toggle (7..)	6 ticks	12t	0 Oil; cuts light cone & sight agro
Brass Loupe Focus	9 ticks	Hold (10..69)	6 ticks	75t	0 Oil; decodes prism & bone-glass glyphs
Phonograph Cancel	24 ticks	Ticks 25..114	36 ticks	150t	Cancels room harmonic; lures Listener
Ferrotypes Flash	18 ticks	Ticks 19..24	66 ticks	90t	Stuns apparitions 150t within 6.5 m
Crouch Sidestep	5 ticks	Ticks 6..16	12 ticks	28t	<= -38 dBFS silent evasion step
Smelling Salts	15 ticks	Ticks 16..45	10 ticks	55t	+40 Composure, -25 Exposure

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / 60 HZ INSTRUMENT TABLE

Deterministic Gate Verification for Diegetic Instrument & Evasion Frame Table

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

THREE STALKER APPARITION ARCHETYPES

Sensory Hunting Rules & Counterplay Contracts

Each apparition archetype hunts through a distinct sensory channel (acoustic dB, refractive line-of-sight, or tidal water ripple) so players must combine all three instruments.

Apparition	Space	Sensory Channel	Speed	Attack / Hazard	Counterplay
Mire Listener	Darkroom / Crypt	Acoustic >= -28 dB	3.4 m/s	Resonance Grasp (-28 Cmp)	Crouch (-38 dB) or Phonograph lure
Glass Septum Watcher	Refraction Gallery	Lantern cone LOS	2.6 m/s	Prism Glare (+35 Exposure)	Shutter lantern; Ferrotypes stun 150t
Drowned Chorister	Submerged Crypt	Water ripple + hum	2.9 m/s	Tidal Undertow (-32 Cmp)	Phonograph Phase-Cancel (ticks 25..114)

Encounter Director Pacing Invariant

Max 2 Active Apparitions

At most 2 apparitions may occupy active hunting states simultaneously; while the player inspects the Stripping Room Evidence Board, exterior apparitions hold patrol nodes.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / APPARITION ARCHETYPES

Deterministic Gate Verification for Sensory Hunting Rules & Counterplay Contracts

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TWO-PHASE HORROR CLIMAX: THE CHOIR IN THE GLASS

Boss Architecture, Phase Transition & Harmonic Vent

Suspended above the flooded nave of The Glass Ossuary, The Choir in the Glass (600 Resonance Integrity) is defeated by acoustic phase cancellation and Ferrotypes exposure rather than conventional weapons.

Phase	Attack Name	Telegraph	Active	Recovery	Effect & Counterplay
P1	Refractive Blind	36 ticks	18 ticks	30 ticks	Sweeping cyan beam; shutter lantern behind pier
P1	Tidal Undertow	42 ticks	24 ticks	36 ticks	Radial water pull; step onto raised basalt plinth
P1	Shatter Harmonic	48 ticks	12 ticks	42 ticks	65 dB acoustic spike; Phonograph Phase-Cancel
P1	Hollow Hymn	54 ticks	30 ticks	120t vent	Opens brass resonator ribs for Ferrotypes Flash (~150 Int)
P2	Prism Split	30 ticks	24 ticks	30 ticks	Splits into 3 refractive decoys; inspect with Brass Loupe
P2	Glass Lung Vacuum	45 ticks	30 ticks	36 ticks	Draws air & oil; shutter lantern + crouch sidestep
P2	Echoing Verdict	36 ticks	18 ticks	30 ticks	Triple ring wave; phase-cancel + plinth elevation
P2	Final Exposure	60 ticks	24 ticks	150t vent	Full organ resonance; Ferrotypes Flash deals 200 Integrity

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / CLIMAX BOSS SPECIFICATION

Deterministic Gate Verification for Boss Architecture, Phase Transition & Harmonic Vent

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

OPTICAL & ACOUSTIC PUZZLE SPECIFICATIONS

Prism Triangulation (Beat 05) & Hydrophone Lock (Beat 06)

Both mid-slice puzzles are deterministic physical simulations integrated into the Three.js scene graph.

Puzzle 01: Prism Triangulation (Refraction Gallery, Beat 05)

3 Concentric Fresnel Rings (45, 135, 270 deg)

Rotate three geared brass Fresnel rings into phase so the tallow lantern beam refracts at 45, 135, and 270 degrees onto the salt-frosted archive vault, revealing the Tide Ledger Fragment.

Puzzle 02: Hydrophone Frequency Triangulation (Submerged Crypt, Beat 06)

3 Sluice Resonators (110 Hz, 220 Hz, 330 Hz)

Adjust three flooded bronze sluice valves until the Wax-Cylinder Phonograph locks the 110/220/330 Hz harmonic triad, unsealing the Hydrophone Cylinder.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / PROCEDURAL PUZZLES

Deterministic Gate Verification for Prism Triangulation (Beat 05) & Hydrophone Lock (Beat 06)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

ZERO-ASSET PROCEDURAL GEOMETRY & TSL SHADERS

Constructive Bone-Glass Ribs, Fresnel Lenses & Lantern Rig

Every mesh in The Glass Ossuary is built from deterministic parametric curves, lathe profiles, and instanced rib grammars.

```
// Procedural Bone-Glass Cathedral Rib Grammar (src/procedural/ossuaryRibs.ts)
export function buildOssuaryRibArch(seed: number, spanMeters = 14.0, heightMeters = 11.5): THREE.BufferGeometry {
  const rng = createXorShift32(seed);
  const curve = new THREE.CatmullRomCurve3([
    new THREE.Vector3(-spanMeters * 0.5, 0, 0),
    new THREE.Vector3(-spanMeters * 0.36, heightMeters * 0.68, 0),
    new THREE.Vector3(0, heightMeters, 0),
    new THREE.Vector3(spanMeters * 0.36, heightMeters * 0.68, 0),
    new THREE.Vector3(spanMeters * 0.5, 0, 0),
  ]);
  return quantizeGeometry1e5(new THREE.TubeGeometry(curve, 48, 0.22 + rng.nextFloat() * 0.04, 12, false));
}
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / PROCEDURAL ART GRAMMAR

Deterministic Gate Verification for Constructive Bone-Glass Ribs, Fresnel Lenses & Lantern Rig

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

PROCEDURAL ACOUSTIC HORROR & EVP SYNTHESIS

Web Audio Graph, Binaural Room Modes & Loudness Gate

All ocean surf, glass-rib sympathetic chimes, wax-cylinder crackle, and Choir organ clusters are synthesized in Web Audio API.

Stem / Cue ID	Synthesis Topology	Loudness & Peak Contract
amb_tide_causeway	Pink noise -> 2-pole bandpass (180 Hz) + LFO swell	-22 LUFS integrated / <= -4.0 dBTP
sfx_wax_phonograph	Formant resonator bank (F1 480, F2 1150, F3 2400 Hz)	-16 LUFS integrated / <= -1.5 dBTP
sfx_ferrottype_flash	Capacitor whine sine sweep (800->6400 Hz) + impulse	-14 LUFS short-term / <= -1.0 dBTP
mus_choir_in_glass	12-voice detuned glass harmonica + sub-bass 55 Hz	-16 LUFS +/- 2 / <= -1.0 dBTP

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / WEB AUDIO SYNTHESIS

Deterministic Gate Verification for Web Audio Graph, Binaural Room Modes & Loudness Gate

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DIEGETIC HUD, EVIDENCE BOARD & ACCESSIBILITY

Minimalist Inspection Overlay & WCAG 2.2 AA Controls

The screen remains clear of arcade bars during exploration; Composure, Lantern Oil, and Exposure are readable on the physical lantern gauge and optional high-contrast HUD.

Visual Acoustic Ring Indicator (Deaf / Hard-of-Hearing Parity)	Radial Edge Compass
Displays directional decibel ticks and harmonic phase-cancel windows visually around the reticle so all audio puzzles and Mire Listener stealth cues are 100% solvable muted.	
Full Input Remapping & Photosensitivity Guard	Ferrotypesafe-Flash Mode
Replaces the white Ferrotypesafe UV flash with a smooth silver-cyan (#4FA89B) iris wipe when Photosensitivity / Reduced-Motion mode is enabled.	

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / DIEGETIC UI & ACCESSIBILITY

Deterministic Gate Verification for Minimalist Inspection Overlay & WCAG 2.2 AA Controls

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

04

DETERMINISTIC REPLAY MATRIX & CRITIC GATES

QA, REPLAYS & CRITICS

Sixteen seeded replay scenarios (rep-01 through rep-16) and calibrated blind critics verify every beat, hypothesis branch, and performance budget.

16 Seeded Replay Scenarios (rep-01..rep-16)

Covers every space, puzzle, apparition counterplay, all three Beat 07 hypotheses, and both Beat 10 endings.

Order-Reversed Blind Critic Pairs

Independent visual, audio, and gameplay critics score artifacts without write access.

SIXTEEN DETERMINISTIC REPLAY SCENARIOS

Canonical Replay Suite (rep-01 through rep-16)

Executed via `npm run test:replay -- --seed=1337`. Each scenario verifies `state_hash`, `audio_hash`, and `geometry_hash`.

Scenario ID	Seed	Ticks	Target Verification & Invariant
rep-01-causeway-boot	1337	600	Boot in <= 2.5s; lantern shutter & crouch dB calibration
rep-02-moreau-cylinder	1337	900	Play Moreau wax cylinder; case_saint_vane == INQUEST_OPENED
rep-03-ferrotype-cipher	1337	1200	Expose salt-print cipher; stun Mire Listener for 150 ticks
rep-04-gallery-stealth	1337	1500	Traverse Refraction Gallery with 0 Septum Watcher alerts
rep-05-prism-triangulation	1337	1800	Align 45/135/270 deg rings; acquire Tide Ledger Fragment
rep-06-crypt-hydrophone	1337	2100	Phase-cancel Drowned Chorister; acquire Hydrophone Cylinder
rep-07-salts-recovery	1337	900	Consume Camphor Smelling Salts; verify +40 Composure
rep-08-oil-replenish	1337	900	Consume Spermaceti Vial; verify +50 Lantern Oil
rep-09-watcher-glare-break	1337	1200	Break Septum Watcher LOS behind basalt pier within 18 ticks
rep-10-chorister-cancel	1337	1200	Phonograph cancel window (ticks 25..114) silences hum
rep-11a-board-lens-sabotage	1337	900	Lock lens_sabotage; verify Loupe +40% & stun 180 ticks
rep-11b-board-tidal-quarantine	1337	900	Lock tidal_quarantine; verify wading -6 dBFS
rep-11c-board-acoustic-calling	1337	900	Lock acoustic_calling; verify cancel window 114 ticks
rep-12-choir-phase1	1337	2400	Reduce The Choir in the Glass to 300 Integrity (Phase 2)
rep-13-choir-phase2-defeat	1337	3000	Defeat The Choir in the Glass; CHOIR_SILENCED
rep-14-verdict-publish	1337	600	Select VERDICT_PUBLISH; verify save persistence
rep-15-verdict-submerge	1337	600	Select VERDICT_SUBMERGE; verify save persistence
rep-16-full-slice-golden	1337	48000	Complete 13.3-min golden run with 0 P0/P1/P2 defects

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / REPLAY MATRIX

Deterministic Gate Verification for Canonical Replay Suite (rep-01 through rep-16)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

PLAYWRIGHT TELEMETRY & INSPECTION CONTRACT

Mandatory Browser Hooks (window.__rpgReady & window.__rpgQA)

The QA harness interacts exclusively through deterministic window hooks so automated gates never rely on fragile pixel sleeps.

```
// Mandatory Global QA Interface for The Glass Ossuary
interface GlassOssuaryQAHarness {
  getState(): OssuaryStateSnapshot;
  getMetrics(): { p50_ms: number; p95_ms: number; p99_ms: number; draw_calls: number; triangles: number };
  stepTicks(count: number): void;
  loadScenario(scenarioId: string, seed: number): Promise<void>;
  captureAudioOffline(durationSeconds: number): Promise<{ lufs: number; dbtp: number; pcm_sha256: string }>;
}
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / BROWSER QA HOOKS

Deterministic Gate Verification for Mandatory Browser Hooks (window.__rpgReady & window.__rpgQA)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

CALIBRATED INDEPENDENT CRITIC GATES

Visual, Audio & Deduction Rubric Thresholds

Independent critics inspect captured PNGs, frame-time histograms, and audio renders in reversed presentation pairs.

Critic Dimension	Pass Threshold	Automatic Defect Trigger
Visual Readability	>= 4.2 / 5.0 across 5 spaces	Luminance floor < 1.8 cd/m2 or off-palette hex
Acoustic Discipline	-16 LUFS +/- 2, <= -1.0 dBTP	Clipping, missing binaural pan, or external audio
Forensic Coherence	100% clue-to-board traceability	Any hypothesis lockable without required proofs
Frame Stability	P95 <= 16.6 ms, P99 <= 22.0 ms	First-flash Ferrottype shader compile spike > 22 ms

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / CRITIC RUBRICS

Deterministic Gate Verification for Visual, Audio & Deduction Rubric Thresholds

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

05

COPY-PASTE MASTER ORCHESTRATOR PROMPT

ORCHESTRATOR PROMPT

Paste this prompt directly into the primary coding orchestrator (or load prompts/glass-ossuary/orchestrator.md from the repository).

Constitutional Authority & Single Write Owner

Enforces the 15-node Evidence Graph, 60 Hz stealth/instrument frame table, and 3-try rollback rule.

EXECUTABLE SYSTEM PROMPT

Master Orchestrator Prompt: Mission & World Contract

Standalone file available at prompts/glass-ossuary/orchestrator.md.

ROLE: You are the Master Orchestrator (`orchestrator`) building *The Glass Ossuary*, a 12-to-16-minute first-person investigative mystery horror vertical slice in Three.js r185 (0.185.0) with zero downloaded runtime assets.

1. FIVE SPACES & TEN-BEAT INQUEST ROUTE (`case\_saint\_vane`):
 - Space 01: Tidewater Causeway (Beat 01 Causeway Landfall)
 - Space 02: Caretaker's Stripping Room (Beat 02 The Sealed Inquest, Beat 07 Deduction)
 - Space 03: Refraction Gallery (Beat 03 Ferrotypes, Beat 04 Watcher, Beat 05 Prism)
 - Space 04: Submerged Crypt (Beat 06 Hydrophone Cylinder vs Drowned Chorister)
 - Space 05: The Glass Ossuary (Beat 08 Unsealing, Beat 09 The Choir in the Glass, Beat 10 Publish or Submerge: VERDICT_PUBLISH | VERDICT_SUBMERGE).
2. PROTAGONIST & DIEGETIC INSTRUMENTS (`Clara Vane`, 100 Composure, 100 Oil, 0-100 Exposure):
 - Split-Diopter Brass Loupe (`brass\_loupe`): 75 ticks focus inspection.
 - Wax-Cylinder Phonograph (`wax\_phonograph`): 24t windup, 90t phase-cancel (25..114).
 - Silver-Salt Ferrotypes Plate (`ferrotypes\_plate`): 18t charge, 6t flash, 150t stun.
3. THREE HYPOTHESES AT BEAT 07 (Stripping Room Evidence Board):
 - `lens\_sabotage` | `tidal\_quarantine` | `acoustic\_calling`.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 05 / MASTER PROMPT (PART 1)

Deterministic Gate Verification for Master Orchestrator Prompt: Mission & World Contract

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

EXECUTABLE SYSTEM PROMPT

Master Orchestrator Prompt: Verification & Rollback Gates

Continuation of prompts/glass-ossuary/orchestrator.md.

- ```
4. FIVE-COLOR PALETTE LOCK:
- #070B10 (Abyssal Tide Slate), #16202A (Salt-Frosted Basalt),
- #C89B54 (Tallow Amber), #4FA89B (Silver-Halide Cyan), #B8423A (Arterial Rust).

5. EVIDENCE GRAPH & DETERMINISM GATES:
- Validate all TaskPacket, DefectRecord, and RunManifest JSON files against schemas/.
- Pre-warm WebGPURenderer via await renderer.compileAsync(scene, camera) and
 await postProcessing.renderAsync() before setting window.__rpgReady = true.
- Run all 16 deterministic replays (`npm run test:replay -- --seed=1337`),
 including rep-11a-board-lens-sabotage, rep-11b-board-tidal-quarantine,
 and rep-11c-board-acoustic-calling.
- Enforce P50 <= 8.3 ms, P95 <= 16.6 ms, P99 <= 22.0 ms, draw calls <= 280,
 triangles <= 460,000, and external_network_requests === 0.
- Maximum 3 repair iterations per node at N13_REPAIR_ROUTER; on non-convergence,
 execute `git reset --hard <accepted_baseline_commit>` and log rollback_triggered.
```

**EVIDENCE GRAPH v2.0 INVARIANT // SECTION 05 / MASTER PROMPT (PART 2)****Deterministic Gate Verification for Master Orchestrator Prompt: Verification & Rollback Gates**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

## 06

SEVEN SPECIALIST SUBAGENT PROMPT CARDS

## SPECIALIST AGENTS

Standalone specialist cards stored under  
prompts/glass-ossuary/agents/\*.md for bounded delegation.

**Six Write-Scoped Builders + One Read-Only Critic**

Each card defines canonical nodes, allowed\_paths, forbidden\_paths, and mechanical acceptance commands.

## SPECIALIST DELEGATION CARDS

# Stealth/Apparitions, World/Inquest & Procedural Art Cards

Standalone files in prompts/glass-ossuary/agents/.

**Agent 01: combat\_gameplay (Stealth, Instruments & Apparitions)****Node:** N06A\_GAMEPLAY\_COMBAT

Implements 60 Hz integer-tick state machine for Clara Vane (Lantern Shutter, Brass Loupe, Phonograph Phase-Cancel, Ferrottype Flash), the 3 apparitions (Mire Listener, Glass Septum Watcher, Drowned Chorister), and The Choir in the Glass.

**Agent 02: world\_quest (Coastal Spaces, Puzzles & Evidence Board)****Node:** N04\_WORLD\_GRAPH

Builds the 5 interconnected spaces, Prism Triangulation (45/135/270 deg), Hydrophone Triangulation (110/220/330 Hz), and the 6-node case\_saint\_vane Evidence Board.

**Agent 03: procedural\_art\_vfx (Bone-Glass Geometry & TSL Shaders)****Nodes:** N03, N05

Generates ribbed basalt vaults, Fresnel lenses, tallow lantern caustics, and Ferrottype silver-halide flash shaders within  $\leq 280$  draw calls and  $\leq 460,000$  triangles.

**EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / AGENT CARDS (1-3)****Deterministic Gate Verification for Stealth/Apparitions, World/Inquest & Procedural Art Cards**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.



SPECIALIST DELEGATION CARDS

# Procedural Audio, UI/A11y, QA/Playwright & Independent Critic

Standalone files in prompts/glass-ossuary/agents/.

**Agent 04: procedural\_audio (Web Audio EVP & Choir Organ)**

Node: N06B\_AUDIO\_SYNTH

Synthesizes coastal storm drones, Caretaker Moreau formant cylinders, Drowned Chorister hums, and 12-voice glass harmonica boss stems (-16 LUFS +/- 2, -1.0 dBTP).

**Agent 05: ui\_hud\_accessibility (Evidence Board & Visual dB Ring)**

Node: N07\_UI\_HUD\_A11Y

Builds the diegetic Evidence Board UI, visual acoustic dB ring for deaf accessibility, full input remapping, and Photosensitivity safe-flash wipe.

**Agent 06: qa\_perf\_playwright (Replays & Telemetry Harness)**

Nodes: N08, N09, N10

Exposes window.\_\_rpgReady / window.\_\_rpgQA and executes rep-01 through rep-16.

**Agent 07: independent\_critic (Blind Rubric & Provenance Audit)**

Nodes: N11, N12

Read-only critic verifying visual luminance floor >= 1.8 cd/m2, -16 LUFS audio, and 0 external requests.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / AGENT CARDS (4-7)

**Deterministic Gate Verification for Procedural Audio, UI/A11y, QA/Playwright & Independent Critic**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

## PACKET DISPATCH &amp; REPAIR ROUTING

# TaskPacket Dispatch & DefectRecord Convergence Protocol

Every specialist handoff is validated against schemas/task-packet.schema.json before N08 static verification runs.

```
Verification Commands Required Before Specialist Handoff
python scripts/verify_release.py
npm run test:perf -- hero-prewarm
npm run test:replay -- --seed=1337
```

## EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / HANDOFF PROTOCOL

**Deterministic Gate Verification for TaskPacket Dispatch & DefectRecord Convergence Protocol**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

## DRAFT 2020-12 JSON CONTRACTS

# TaskPacket, DefectRecord & RunManifest Contracts

Validated against schemas/\*.schema.json and golden fixtures in examples/run-0002/.

```
// Golden Fixture Excerpt: examples/run-0002/run-manifest.json
{
 "run_id": "run-0002",
 "git_sha": "7934f46",
 "three_version": "0.185.0",
 "determinism_regime": "A-PinnedBrowser",
 "backend": "webgpu",
 "seed_table": { "world": 2401, "deduction": 2402, "audio": 2403 },
 "scenarios": [
 { "id": "rep-01-causeway-boot", "seed": 2401, "ticks": 600, "verdict": "passed" },
 { "id": "rep-11a-board-lens-sabotage", "seed": 2402, "ticks": 900, "verdict": "passed" },
 { "id": "rep-13-choir-phase2-defeat", "seed": 2401, "ticks": 3000, "verdict": "passed" }
],
 "metrics": { "frame_time_p50_ms": 7.4, "frame_time_p95_ms": 13.8, "frame_time_p99_ms": 17.6, "external_asset_requests": 0 }
}
```

## EVIDENCE GRAPH v2.0 INVARIANT // APPENDIX A / NORMATIVE SCHEMAS

**Deterministic Gate Verification for TaskPacket, DefectRecord & RunManifest Contracts**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

## NORMATIVE STANDARDS, RUNTIME SPECIFICATIONS &amp; PRIMARY SOURCES

## Primary Technical References &amp; Live URI Registry

- [01] **Three.js r185 (0.185.0) Release Notes & Migration Guide**  
<https://github.com/mrdoob/three.js/releases/tag/r185>
- [02] **Three.js WebGPURenderer API & NodeMaterial Documentation**  
<https://threejs.org/docs/#api/en/renderers/webgpu/WebGPURenderer>
- [03] **Three.js Shading Language (TSL) Specification & Wiki**  
<https://github.com/mrdoob/three.js/wiki/Three.js-Shading-Language>
- [04] **W3C WebGPU Specification (Working Draft)**  
<https://www.w3.org/TR/webgpu/>
- [05] **W3C WebGPU Shading Language (WGSL)**  
<https://www.w3.org/TR/WGSL/>
- [06] **Khronos WebGL 2.0 Specification**  
<https://registry.khronos.org/webgl/specs/latest/2.0/>
- [07] **W3C Web Audio API 1.1 & OfflineAudioContext Specification**  
<https://www.w3.org/TR/webaudio-1.1/>
- [08] **ITU-R BS.1770-5 Programme Loudness & True-Peak Audio Measurement**  
<https://www.itu.int/rec/R-REC-BS.1770>
- [09] **EBU Tech 3341 Loudness Metering: EBU Mode (-16 LUFS / -1.0 dBTP)**  
<https://tech.ebu.ch/publications/tech3341>
- [10] **JSON Schema Draft 2020-12 Core & Validation Specifications**  
<https://json-schema.org/draft/2020-12/schema>
- [11] **Playwright Browser Automation & Deterministic Clock Harness**  
<https://playwright.dev/docs/clock>
- [12] **W3C WCAG 2.2 Accessibility Guidelines (Contrast, Motion, Captions)**  
<https://www.w3.org/TR/WCAG22/>
- [13] **Pointer Lock 2.0 W3C Specification for First-Person Camera Input**  
<https://www.w3.org/TR/pointerlock-2/>
- [14] **Gamepad API W3C Specification for Controller Remapping**  
<https://www.w3.org/TR/gamepad/>
- [15] **Three.js InstancedMesh & BatchedMesh Draw-Call Budgeting**  
<https://threejs.org/docs/#api/en/objects/InstancedMesh>
- [16] **Three.js PMREMGGenerator Procedural Environment Lighting**  
<https://threejs.org/docs/#api/en/extras/PMREMGGenerator>
- [17] **IEEE 754-2019 Floating-Point Arithmetic & Quantization Boundaries**  
<https://standards.ieee.org/ieee/754/6210/>
- [18] **Citation File Format (CFF 1.2.0) Schema Specification**  
<https://citation-file-format.github.io/>
- [19] **Three.js Evidence Graph & Multi-Genre Game Archive Repository**  
<https://github.com/Emily2040/-threejs-evidence-graph>
- [20] **Three.js PostProcessing & Pass Node Graph Architecture**  
<https://github.com/mrdoob/three.js/tree/r185/examples/jsm/tsl>